

MATH 345 Skeleton Notes

Unit 3: Beyond linear maps: local linearization

Section 3.2: Balls, limits, and continuity

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Limits and continuity of scalar-valued functions

Balls, limits, and continuity

This short section supplies local language used in derivative and optimization statements: balls, limits, continuity, interior points, boundary points, closed sets, and bounded sets.

Recall the definition of a limit from single-variable calculus. For a function $f : \mathbb{R} \rightarrow \mathbb{R}$ and $a, b \in \mathbb{R}$, the equation

$$\lim_{x \rightarrow a} f(x) = b$$

holds if for each $\varepsilon > 0$, there is $\delta > 0$ so that if $0 < |x - a| < \delta$, then $|f(x) - b| < \varepsilon$. We can interpret this statement geometrically as follows: for any open interval J centered at b , there is an open interval I centered at a so that f maps points in I , excluding a itself, into J .

In $\mathbb{R}^2$, a ball is a disk.

Definition: Disks in $\mathbb{R}^2$ Consider a point $(a, b) \in \mathbb{R}^2$. The δ -disk centered at point (a, b) is the region of $\mathbb{R}^2$ bounded by the circle of radius δ centered at (a, b) , i.e., the set

$$\{(x, y) \in \mathbb{R}^2 : \sqrt{(x - a)^2 + (y - b)^2} < \delta\}.$$

See textbook figure fig-8-4-delta-disk for a depiction of a δ disk.

We now define the limit of a scalar-valued function f of two variables. For this definition, the equation

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L,$$

will then hold precisely when, for each open interval I centered at L , there is a δ disk centered at (a,b) so that all points in this disk which are also contained in the domain of f , excluding the point (a,b) itself, are mapped by f into I .

Definition: Limits of Scalar-Valued Functions of Two Variables

Consider a scalar-valued function $f(x,y)$ of two variables. The of f as (x,y) approaches (a,b) is equal to L , written

limit

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L,$$

if for each $\varepsilon > 0$, there is $\delta > 0$ so that for any points in the domain of f with

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta,$$

the inequality $|f(x,y) - L| < \varepsilon$ holds.

Using vector-notation (recalling Length of a vector), we can write a δ -disk centered at a point $\mathbf{x}_0 \in \mathbb{R}^2$ as the set of points $\mathbf{x} \in \mathbb{R}^2$ so that

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x} - \mathbf{x}_0\| < \delta\}.$$

This motivates the higher dimensional variant of a δ disk: a δ -ball.

Definition: Balls in $\mathbb{R}^n$ For $\mathbf{x}_0 \in \mathbb{R}^n$, the δ -ball centered at $\mathbf{x}_0$ is the set

δ -ball

$$\{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{x}_0\| < \delta\}.$$

In $\mathbb{R}^3$ a δ -ball centered at a point (a, b, c) is precisely the set of points satisfying the equation

$$(x - a)^2 + (y - b)^2 + (z - c)^2 < \delta^2.$$

It is the region in $\mathbb{R}^3$ bounded by a sphere of radius δ .

By replacing disks with balls, we obtain the definition of a limit of a general scalar-valued function.

Definition: The Limit of a Scalar-Valued Function Consider a scalar-valued function f . The limit of f as $\mathbf{x}$ approaches $\mathbf{x}_0$ is equal to L , written

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} f(\mathbf{x}) = L,$$

if for each $\varepsilon > 0$, there is $\delta > 0$ so that for any points in the domain of f with $0 < \|\mathbf{x} - \mathbf{x}_0\| < \delta$, the inequality $|f(\mathbf{x}) - L| < \varepsilon$ holds.

Remark In order for the limit of f , when it exists, to be unique, the point $\mathbf{x}_0$ must have the property that the domain of f contains a point other than $\mathbf{x}_0$ in any δ -ball centered at $\mathbf{x}_0$. This is only a small issue, since every point in the domain of any of the functions that naturally occur in calculus have this property.

Recall from the calculus of a single variable that a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at a point a when $\lim_{x \rightarrow a} f(x) = f(a)$, i.e., when the limit on the left hand side exists and is equal to the right hand side. The continuity of functions of several variables is defined in exactly the same way.

Definition A scalar-valued function f is continuous at a point $\mathbf{x}_0$ in its domain if the limit of f as $\mathbf{x} \rightarrow \mathbf{x}_0$ exists, and

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} f(\mathbf{x}) = f(\mathbf{x}_0).$$

Interior, boundary, and closed sets

This language lets us distinguish points inside a set from points on its boundary.

Definition: Interior, boundary, and closed sets Let S be a sub-

set of $\mathbb{R}^n$. Then a point $\mathbf{x}_0 \in S$ is an **interior** point if there is a δ -ball centered at $\mathbf{x}_0$ completely contained in S . A point $\mathbf{x}_0 \in \mathbb{R}^n$ is a

boundary point if every ball centered at $\mathbf{x}_0$ contains at least one point of S and at least one point outside S . The set of all boundary

points of a set S is called the **boundary** of the set S .

Consider the figure for an illustration of the example of an interior point and boundary point.

Remark Note that a boundary point of a set S *need not be contained in the set* S . For example, the δ -disk defined in this section does not contain its circular boundary. A set that contains its boundary is called *closed*.

Definition: Closed sets A set $S \subset \mathbb{R}^n$ is called **closed** if it

contains its boundary. A set $S \subset \mathbb{R}^n$ is **bounded** if it is contained in some ball centered at the origin, or equivalently if all points of S lie within some fixed distance of the origin.